Chapter 9. Building and Extending Loglinear–Logit Models

Collapsibility, ordinal association, rates, sparseness

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This chapter turns the loglinear models of Chapter 8 into a working toolkit: association graphs that display conditional independence and tell us when a table may be collapsed (Section 9.1), model selection and comparison (Section 9.2), residual diagnostics (Section 9.3), models that use the ordering of categories (Section 9.4), Poisson regression for rates (Section 9.5), and the problems caused by empty cells and sparse tables (Section 9.6).

9.1 Association graphs and collapsibility

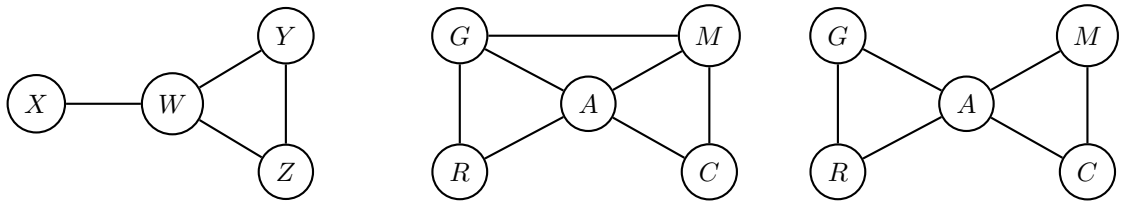
Motivation. With four or five variables it is hard to read conditional independence off a model formula. Drawing the model as a graph turns these questions into questions about paths, and it also tells us when a table can be collapsed over some variables without changing the associations of interest.

9.1.1 Association graphs

Definition (Association graph, path, separation)

The *association graph* of a loglinear model has a vertex for each variable and an edge between two variables if and only if the model contains a term involving both, i.e. if they are conditionally associated. A *path* is a sequence of edges leading from one variable to another. A set S of variables *separates* X and Y if every path between X and Y passes through a vertex of S.

Models with the same pairwise associations have the same graph; e.g. (WX, WY, WZ, YZ) and (WX, WYZ) share Figure 1.



(WX, WY, WZ, YZ) model 6: $(AC, AM, CM, AG, AR, GM, GR)$ model 7: (AC, AM, CM, AG, AR, GR)

Figure 1: Association graphs. Left: book Figure 9.1; W separates X from $\{Y, Z\}$. Middle and right: two models for the five-way student survey (Example 9.5); in model 7, A separates $\{G, R\}$ from $\{C, M\}$.

Theorem 9.1 (Separation implies conditional independence)

In a hierarchical loglinear model, if a set S of variables separates X and Y in the association graph, then X and Y are conditionally independent given S (after marginalizing over all other variables). More generally, if S separates sets A_0 and B_0, then $A_0 \perp B_0 \mid S$.

Proof. [Supplement]

Step 1. Let V be the set of all variables, A the set of vertices reachable from X by paths avoiding S, and $B = V \setminus (A \cup S)$. Since S separates X and Y, $Y \in B$.

Step 2. Every term T of the model lies in $A \cup S$ or in $B \cup S$. Indeed, if T contained $a \in A$ and $b \in B$, then the hierarchical model would contain the two-factor term $\{a, b\} \subseteq T$, so a–b would be an edge and b would be reachable from X avoiding S, i.e. $b \in A$, a contradiction.

Step 3. Grouping terms accordingly, $\mu(\mathbf{x}) = f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)$ (assign terms inside S to f). Hence, for fixed $\mathbf{x}_S$,

$$P(\mathbf{x}_A, \mathbf{x}_B \mid \mathbf{x}_S) = \frac{f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)}{\sum_{\mathbf{a}} f(\mathbf{a}, \mathbf{x}_S) \sum_{\mathbf{b}} g(\mathbf{b}, \mathbf{x}_S)},$$

which factors into a function of $\mathbf{x}_A$ times a function of $\mathbf{x}_B$: $\mathbf{x}_A \perp \mathbf{x}_B \mid \mathbf{x}_S$.

Step 4. Summing over the coordinates of A other than X and the coordinates of B other than Y preserves the product form, so $X \perp Y \mid S$. The statement for sets $A_0 \subseteq A$, $B_0 \subseteq B$ is the same computation. ■

In Figure 1 (left), W separates X and Y, so $X \perp Y \mid W$ (and also given $\{W, Z\}$); likewise $X \perp Z \mid W$.

9.1.2 Collapsibility in three-way contingency tables

Theorem 9.2 (Collapsibility conditions)

In a three-way table, the XY marginal odds ratios equal the XY conditional odds ratios (at every level of Z) if Z and X are conditionally independent given Y, or Z and Y are conditionally independent given X; that is, under models (XY, YZ) or (XY, XZ) (or simpler models).

Proof. [Supplement]

Step 1. Under (XY, XZ), by Lemma 8.5, $\mu_{ijk} = A_{ij}C_{ik}$. The conditional odds ratio at level k is

$$\frac{A_{ij}C_{ik}\, A_{i'j'}C_{i'k}}{A_{ij'}C_{ik}\, A_{i'j}C_{i'k}} = \frac{A_{ij}A_{i'j'}}{A_{ij'}A_{i'j}},$$

free of k. The marginal table is $\mu_{ij+} = A_{ij}C_{i+}$, and the row factors $C_{i+}, C_{i'+}$ cancel in the odds ratio, giving the same value.

Step 2. Under (XY, YZ), $\mu_{ijk} = A_{ij}B_{jk}$ and $\mu_{ij+} = A_{ij}B_{j+}$; now column factors cancel, with the same conclusion. ■

Example 9.3 (Collapsibility in the student survey)

In (AM, CM) (graph A–M–C), C is conditionally independent of A, so the AM association is collapsible over C: fitted conditional and marginal AM odds ratios both equal 61.9 (Example 8.8); similarly CM (25.1). The AC association is not collapsible, because M is linked to both A and C: conditional AC odds ratio 1.0, marginal 2.7. In (AC, AM, CM) no collapsibility condition holds, and every pair has quite different conditional and marginal associations.

9.1.3 Collapsibility and logit models

In a multicenter clinical trial with binary treatment x, binary response Y and centers Z, the model $\operatorname{logit} P(Y = 1 \mid X = i, Z = k) = \alpha + \beta x_i + \beta_k^Z$ corresponds to (XY, XZ, YZ), so the common conditional odds ratio e^{β} typically differs from the odds ratio of the marginal 2×2 table. The simpler model without center effects, $\operatorname{logit} = \alpha + \beta x_i$, states $Y \perp Z \mid X$; it is equivalent to (XY, XZ), which is collapsible over Z. So when center effects are negligible the treatment effect is approximately the marginal odds ratio.

9.1.4 Collapsibility and association graphs for multiway tables

Theorem 9.4 (Parametric collapsibility, Bishop et al.)

Partition the variables into disjoint sets A, B, C such that B separates A and C in the association graph. After collapsing the table over the variables in C, the parameters relating variables within A, and those relating variables in A to variables in B, are unchanged.

Proof. [Supplement]

Step 1. No term contains variables from both A and C (that would create an A–C edge, a path avoiding B). Put all terms meeting A into f and all others into g; then each term of f lies in $A \cup B$ and each term of g lies in $B \cup C$:

$$\mu(\mathbf{a}, \mathbf{b}, \mathbf{c}) = f(\mathbf{a}, \mathbf{b})\, g(\mathbf{b}, \mathbf{c}), \qquad \log f = \sum_{T \cap A \neq \emptyset} \lambda^T.$$

Step 2. Collapsing over C: $\mu(\mathbf{a}, \mathbf{b}) = f(\mathbf{a}, \mathbf{b})\, G(\mathbf{b})$ with $G(\mathbf{b}) = \sum_{\mathbf{c}} g(\mathbf{b}, \mathbf{c})$. Thus

$$\log \mu(\mathbf{a}, \mathbf{b}) = \sum_{T \cap A \neq \emptyset} \lambda^T + \log G(\mathbf{b}).$$

Step 3. $\log G(\mathbf{b})$ decomposes into terms involving only variables of B. Since the decomposition of $\log \mu(\mathbf{a}, \mathbf{b})$ into main-effect and interaction terms is unique under fixed (e.g. baseline) constraints, every term involving a variable of A in the collapsed table equals the corresponding λ^T of the full model. ■

For (WX, WY, WZ, YZ): with $A = \{X\}$, $B = \{W\}$, $C = \{Y, Z\}$, the WX association is unchanged after collapsing over Y, Z; with $A = \{Y, Z\}$, $B = \{W\}$, $C = \{X\}$, the associations among W, Y, Z are unchanged after collapsing over X. A variable independent of all others can be collapsed over without affecting other terms. When B has several variables the parameters are unchanged but their ML estimates may differ slightly; estimates also agree ("commutativity of fitting and collapsing") if the model contains the highest-order term relating the variables of B to each other.

9.2 Model selection and comparison

Motivation. Even with five variables there are thousands of hierarchical loglinear models. We need principles (which terms must be included) and tools (likelihood-ratio comparisons of nested models) to find a model that fits well but is simple enough to interpret.

9.2.1 Considerations in model selection

A model should be complex enough to fit well yet simple to interpret, smoothing rather than overfitting. Guidelines:

(i) Distinguish responses from explanatory variables and concentrate on terms linking responses and linking explanatory variables to responses.

(ii) Include the most general interaction among the explanatory variables, and any margin fixed by the sampling design; then the likelihood equations reproduce these totals. For Table 8.8 with responses I, S and predictors G, L, the model should contain GL (so the fit has 20,629 urban women as observed), i.e. be at least as complex as (GL, S, I).

(iii) With a single categorical response, use the equivalent logit model.

(iv) For exploration, fit the model with all single-factor terms, then all two-factor terms, then all three-factor terms, and so on, to locate the needed complexity; automatic searches (backward elimination) should be used with care.

9.2.2 Model building for the Dayton student survey

Example 9.5 (Five-way student survey)

The survey of Example 8.8 also recorded gender (G) and race (R):

		Race = white				Race = other			
		Female		Male		Female		Male	
A	C	M=yes	no	yes	no	yes	no	yes	no
yes	yes	405	268	453	228	23	23	30	19
yes	no	13	218	28	201	2	19	1	18
no	yes	1	17	1	17	0	1	1	8
no	no	1	117	1	133	0	12	0	17

Treating A, C, M as responses and G, R as explanatory, every model contains GR.

Model	G^2	df
1. Mutual independence $+ GR$	1325.1	25
2. Homogeneous association (all two-factor terms)	15.3	16
3. All three-factor terms	5.3	6
4g. (2) $- CR$	15.8	17
5. $(AC, AM, CM, AG, AR, GM, GR, MR)$	16.7	18
6. $(AC, AM, CM, AG, AR, GM, GR)$	19.9	19
7. (AC, AM, CM, AG, AR, GR)	28.8	20

Model 3 improves on model 2 only by $15.3 - 5.3 = 10.0$ on 10 df, so three-factor terms are dropped. Backward elimination from model 2 (removing each of the 9 candidate pairs, models 4a–4i, e.g. removing AC gives 201.2 and removing CM gives 513.5) first removes CR (increase 0.5, df 1), then CG (+0.9), then MR (+3.2); removing AG next would add 5.3 ($P = 0.02$), so we stop at model 6. (These P-values are not literal, since the data chose the tests; many counts are small, but G^2 differences remain useful.)

In model 6 (Figure 1, middle) every path from C to $\{G, R\}$ goes through $\{A, M\}$: given alcohol and marijuana use, cigarette use is independent of gender and race. By Theorem 9.4, the AC and CM conditional associations equal those of (AC, AM, CM) fitted to the ACM table collapsed over G, R. Model 7 drops GM; then A separates $\{G, R\}$ from $\{C, M\}$, so all associations among A, C, M equal those of (AC, AM, CM) in the collapsed table. Model 7 fits reasonably ($G^2 = 28.8$, df $= 20$, $\hat{\Delta} = 0.036$), so one may collapse over gender and race to study A, C, M; the full model additionally estimates the effects of G and R on the responses.

9.2.3 Loglinear model comparison statistics

Let M_0 be a special case of M_1, with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$.

Theorem 9.6 (Partitioning G^2)

$$G^2(M_0) = G^2(M_1) + G^2(M_0 \mid M_1), \tag{9.1}$$

where the likelihood-ratio statistic for M_0 against M_1 has the form

$$G^2(M_0 \mid M_1) = G^2(M_0) - G^2(M_1) = 2\sum_i \hat{\mu}_{1i} \log \frac{\hat{\mu}_{1i}}{\hat{\mu}_{0i}}, \tag{9.2}$$

which is the G^2 statistic for M_0 with $\hat{\boldsymbol{\mu}}_1$ playing the role of the data.

Proof.

Step 1. $G^2(M) = 2\sum n_i \log(n_i/\hat{\mu}_i)$, so

$$G^2(M_0) - G^2(M_1) = 2\sum_i n_i(\log \hat{\mu}_{1i} - \log \hat{\mu}_{0i}).$$

Step 2. Write $\log \hat{\boldsymbol{\mu}}_1 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_1$ and, since $M_0 \subseteq M_1$, $\log \hat{\boldsymbol{\mu}}_0 = \mathbf{X}_0\hat{\boldsymbol{\beta}}_0 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_0^*$, where $\hat{\boldsymbol{\beta}}_0^*$ is $\hat{\boldsymbol{\beta}}_0$ with zeros appended for the extra parameters. Then

$$2\mathbf{n}'(\log \hat{\boldsymbol{\mu}}_1 - \log \hat{\boldsymbol{\mu}}_0) = 2\,\mathbf{n}'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*) = 2\,\hat{\boldsymbol{\mu}}_1'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*),$$

using the likelihood equations $\mathbf{X}_1'\mathbf{n} = \mathbf{X}_1'\hat{\boldsymbol{\mu}}_1$ for M_1 (Theorem 8.18). This is $2\sum \hat{\mu}_{1i}\log(\hat{\mu}_{1i}/\hat{\mu}_{0i})$.

Step 3. $G^2(M_0 \mid M_1)$ has the G^2 form. (Note also $\sum \hat{\mu}_{0i} = \sum \hat{\mu}_{1i} = n$ when both models contain an intercept.) Equation (9.1) is the definition rearranged. ■

$G^2(M_0 \mid M_1)$ depends on the data only through $\hat{\boldsymbol{\mu}}_1$, i.e. through the sufficient statistics of M_1. The Pearson difference $X^2(M_0) - X^2(M_1)$ need not be nonnegative; the natural Pearson analogue is

$$X^2(M_0 \mid M_1) = \sum_i \frac{(\hat{\mu}_{1i} - \hat{\mu}_{0i})^2}{\hat{\mu}_{0i}}. \tag{9.3}$$

When M_0 holds, both are asymptotically χ^2 with $\mathrm{df} = \mathrm{df}(M_0) - \mathrm{df}(M_1)$, and they behave similarly even in fairly sparse tables (Haberman). When M_1 holds but M_0 does not, $G^2(M_1)$ remains χ^2 while the others grow without bound.

9.2.4 Partitioning chi-squared with model comparisons

Applying (9.1) repeatedly to a nested sequence $M_1 \supset M_2 \supset \cdots$ partitions G^2 into components that are asymptotically independent. Example: for a $2 \times J$ table let M_j state that columns $1, \ldots, j$ form an independence table. Then

$$G^2(M_J) = G^2(M_2) + G^2(M_3 \mid M_2) + \cdots + G^2(M_J \mid M_{J-1}),$$

and substituting the fitted values shows that $G^2(M_j \mid M_{j-1})$ is the G^2 for independence in the 2×2 table whose first column combines columns $1, \ldots, j-1$ and whose second column is column j. This justifies the partitioning of Section 3.3.3.

With s planned tests for nested models, using level $1-(1-\alpha)^{1/s}$ for each keeps the overall asymptotic type I error at most α (the tests being asymptotically independent). For $s = 3$ and $\alpha = 0.05$, use $1 - 0.95^{1/3} = 0.017$ per test.

9.2.5 Identical marginal and conditional tests of independence

When both models are direct, they contain the independence linkages that guarantee collapsibility, and the conditional test reduces to a marginal one (Sundberg, 1975): if two direct models differ

only by one pairwise term, $G^2(M_0 \mid M_1)$ equals G^2 for independence in the marginal table of that pair.

Proposition 9.7 (A conditional test that equals a marginal test)

$G^2[(X,Y,Z) \mid (XY,Z)]$ equals $G^2(X,Y)$ for testing independence in the marginal XY table.

Proof. By Theorem 8.20, $\hat{\mu}_{1,ijk} = n_{ij+}n_{++k}/n$ and $\hat{\mu}_{0,ijk} = n_{i++}n_{+j+}n_{++k}/n^2$, so $\hat{\mu}_1/\hat{\mu}_0 = n\,n_{ij+}/(n_{i++}n_{+j+})$ does not depend on k. By (9.2),

$$G^2 = 2\sum_{i,j,k} \frac{n_{ij+}n_{++k}}{n} \log \frac{n\,n_{ij+}}{n_{i++}n_{+j+}} = 2\sum_{i,j} n_{ij+} \log \frac{n_{ij+}}{n_{i++}n_{+j+}/n},$$

after summing n_{++k}/n over k. This is G^2 for independence in the $\{n_{ij+}\}$ table. ■

This is expected: under (XY,Z) the XY association is collapsible over Z.

9.3 Diagnostics for checking models

Motivation. G^2 tells us whether a model fits; residuals tell us where it fails, cell by cell.

9.3.1 Residuals for loglinear models

Definition (Pearson and standardized residuals)

For cell i the *Pearson residual* is

$$e_i = \frac{n_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}, \qquad \sum_i e_i^2 = X^2. \tag{9.4}$$

The *standardized Pearson residual* is $r_i = e_i/\sqrt{1-\hat{h}_i}$, where $\hat{h}_i$ is the leverage (diagonal of the estimated hat matrix).

Pearson residuals have asymptotic variances below 1 (on average residual df/number of cells); r_i is asymptotically $N(0,1)$ and is preferable. Deviance residuals are an alternative.

9.3.2 Student survey example revisited

For the five-way survey with all two-factor terms, the only large standardized residual is 3.2 (fitted 3.1 for a count of 8); for model 6 it is 3.3 (fitted 2.9) in the same cell: nonwhite males who smoked cigarettes but used neither alcohol nor marijuana. Given the many cells and large n, neither model looks problematic.

9.3.3 Correspondence between loglinear and logit residuals

Proposition 9.8 (Logit versus loglinear Pearson residuals)

For equivalent logit and loglinear models with binary response, the logit-model Pearson residual for a success count equals the loglinear Pearson residual for that cell divided by $\sqrt{1-\hat{\pi}}$; hence it is larger in absolute value.

Proof. Let the cell have count y out of N trials in its binomial row. The fitted values agree: $\hat{\mu} = N\hat{\pi}$ (Theorem 8.16). The logit residual is $(y - N\hat{\pi})/\sqrt{N\hat{\pi}(1-\hat{\pi})}$ and the loglinear residual is $(y-\hat{\mu})/\sqrt{\hat{\mu}} = (y - N\hat{\pi})/\sqrt{N\hat{\pi}}$. Their ratio is $1/\sqrt{1-\hat{\pi}} > 1$. ■

After standardization by their own estimated standard errors, the two kinds of residual are identical, another reason to prefer standardized residuals.

9.4 Modeling ordinal associations

Motivation. Nominal loglinear models ignore category order: permuting rows gives the same fit. For two ordinal variables the saturated model is useless and independence often fails mostly in the corner cells; a single "trend" parameter captures this.

Example 9.9 (Premarital sex and teenage birth control)

1991 General Social Survey. Rows: premarital sex is always wrong (1), almost always wrong (2), wrong only sometimes (3), not wrong at all (4). Columns: teenage birth control should be available, strongly disagree (1) to strongly agree (4). Entries: count, (independence fit), [standardized residual], {L×L fit}.

	SD	D	A	SA
1	81 (42.4) [7.6] {80.9}	68 (51.2) [3.1] {67.6}	60 (86.4) [−4.1] {69.4}	38 (67.0) [−4.8] {29.1}
2	24 (16.0) [2.3] {20.8}	26 (19.3) [1.8] {23.1}	29 (32.5) [−0.8] {31.5}	14 (25.2) [−2.8] {17.6}
3	18 (30.1) [−2.7] {24.4}	41 (36.3) [1.0] {36.1}	74 (61.2) [2.2] {65.7}	42 (47.4) [−1.0] {48.8}
4	36 (70.6) [−6.1] {33.0}	57 (85.2) [−4.6] {65.1}	161 (143.8) [2.4] {157.4}	157 (111.4) [6.8] {155.5}

Independence fits poorly, $G^2(I) = 127.6$, df $= 9$; the corner residuals show a positive trend.

9.4.1 Linear-by-linear association in two-way tables

Definition (Linear-by-linear association (L×L) model)

With ordered row scores $u_1 \le \cdots \le u_I$ and column scores $v_1 \le \cdots \le v_J$,

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \beta u_i v_j, \tag{9.5}$$

the special case $\lambda_{ij}^{XY} = \beta u_i v_j$ of the saturated model: one association parameter instead of $(I-1)(J-1)$. Independence is $\beta = 0$. With $u_i = i$, $v_j = j$ it is the *uniform association* model.

Theorem 9.10 (Odds ratios under L×L)

Under (9.5), for rows $a < c$ and columns $b < d$,

$$\log \frac{\mu_{ab}\mu_{cd}}{\mu_{ad}\mu_{cb}} = \beta(u_c - u_a)(v_d - v_b). \tag{9.6}$$

With unit-spaced scores all local odds ratios equal e^{β}. If $\beta > 0$, then Y tends to increase with X and μ_{ij} exceeds its independence value where X, Y are both high or both low.

Proof.

Step 1. By Theorem 8.3(b) the log odds ratio equals $\lambda_{ab}^{XY} + \lambda_{cd}^{XY} - \lambda_{ad}^{XY} - \lambda_{cb}^{XY}$ with $\lambda_{ij}^{XY} = \beta u_i v_j$:

$$\beta(u_a v_b + u_c v_d - u_a v_d - u_c v_b) = \beta(u_c - u_a)(v_d - v_b).$$

Step 2. For adjacent rows and columns with unit spacing, $(u_{i+1} - u_i)(v_{j+1} - v_j) = 1$.

Step 3. For $\beta > 0$ all odds ratios with $a < c$, $b < d$ exceed 1, which is positive (likelihood-ratio type) dependence. The deviation $\beta u_i v_j$ from independence is largest in absolute value at the corners. ■

The deviation $\beta u_i v_j$ is linear in the column scores within a row (slope βu_i) and linear in the row scores within a column, hence the name. Scores are often midpoints of an underlying continuous scale, or standardized (mean 0, variance 1 under the margins); then β is a log odds ratio for unit standard-deviation changes, and for an underlying bivariate normal with correlation ρ, $\beta \approx \rho/(1-\rho^2)$ ($\approx \rho$ for weak association).

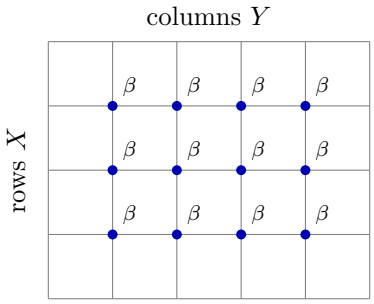


Figure 2: Uniform association: every 2×2 table of adjacent rows and columns has log odds ratio β (book Figure 9.2).

9.4.2 Corresponding logit model for adjacent responses

Proposition 9.11 (Adjacent-categories logit form)

Let $\pi_{j|i} = P(Y = j \mid X = i)$. Under (9.5),

$$\log \frac{\pi_{j+1|i}}{\pi_{j|i}} = (\lambda_{j+1}^{Y} - \lambda_{j}^{Y}) + \beta(v_{j+1} - v_j)u_i,$$

and for unit-spaced $\{v_j\}$ this is $\alpha_j + \beta u_i$: the same linear effect of X on all $J-1$ adjacent-categories logits.

Proof. $\pi_{j+1|i}/\pi_{j|i} = \mu_{i,j+1}/\mu_{ij}$; subtracting (9.5) at j from (9.5) at $j+1$, λ and λ_i^X cancel, leaving the stated expression. ■

9.4.3 Likelihood equations and model fitting

Theorem 9.12 (Likelihood equations for L×L)

The ML fit of (9.5) satisfies

$$\hat{\mu}_{i+} = n_{i+}, \qquad \hat{\mu}_{+j} = n_{+j}, \qquad \sum_{i,j} u_i v_j \hat{\mu}_{ij} = \sum_{i,j} u_i v_j n_{ij}.$$

Hence the fitted and observed distributions have the same marginal means and variances of the scores and the same correlation between them. The residual df are $IJ - I - J$.

Proof.

Step 1. The Poisson log likelihood is

$$L = n\lambda + \sum_i n_{i+}\lambda_i^X + \sum_j n_{+j}\lambda_j^Y + \beta \sum_{i,j} u_i v_j n_{ij} - \sum_{i,j} \exp(\lambda + \lambda_i^X + \lambda_j^Y + \beta u_i v_j).$$

Differentiating with respect to λ_i^X, λ_j^Y and β and equating to zero gives the three sets of equations (the sufficient statistics are the margins and $\sum u_i v_j n_{ij}$).

Step 2. Dividing by n, the fitted proportions $\hat{\pi}_{ij}$ and the sample proportions p_{ij} have equal margins, hence equal means and variances of u and of v, and equal $\sum u_i v_j \hat{\pi}_{ij} = \sum u_i v_j p_{ij}$, hence equal covariance and correlation.

Step 3. The model has $1 + (I-1) + (J-1) + 1$ parameters, so df $= IJ - (I + J)$, unsaturated except for 2×2 tables. ■

The fitted counts therefore reproduce the trend of the data. Fitting uses Newton–Raphson.

9.4.4 Sex opinions example

Example 9.13 (L×L fit to Example 9.9)

With scores $1, 2, 3, 4$ for both variables, $G^2(\text{L}\times\text{L}) = 11.5$, df $= 8$, a dramatic improvement over 127.6 on 9 df, especially in the corners. $\hat{\beta} = 0.286$ (SE 0.028): local odds ratio $e^{0.286} = 1.33$, 95% CI $\exp(0.286 \pm 1.96 \times 0.028) = (1.26, 1.41)$. Nonlocal odds ratios are larger: for the four corner cells, by (9.6),

$$\exp[0.286(4-1)(4-1)] = 13.1 = \frac{80.9 \times 155.5}{29.1 \times 33.0}.$$

Rescaled equally spaced scores (e.g. rows $2, 4, 6, 8$) give the same fit with $\hat{\beta}$ rescaled (0.143). Scores $1, 2, 4, 5$ (treating categories 2 and 3 as farther apart) give $G^2 = 8.8$, $\hat{\beta} = 0.146$. Standardized scores give $\hat{\beta} = 0.374$, and solving $\beta = \rho/(1-\rho^2)$ gives $\hat{\rho} = 0.333$ for an underlying bivariate normal.

9.4.5 Directed ordinal test of independence

Within L×L, independence is $H_0 : \beta = 0$, tested by

$$G^2(I \mid \text{L} \times \text{L}) = G^2(I) - G^2(\text{L} \times \text{L}), \qquad \text{df} = 1.$$

For Example 9.9: $127.6 - 11.5 = 116.1$; the Wald statistic $(\hat{\beta}/\text{SE})^2 = 102.5$; the score statistic is the correlation statistic $M^2 = (n-1)r^2 = 112.6$ of Section 3.4.1.

Why the ordinal test is more powerful

When L×L holds, $G^2(I \mid \mathrm{L}\times\mathrm{L})$ and $G^2(I)$ have (asymptotically) the same noncentrality, but the former has df $= 1$ instead of $(I-1)(J-1)$. For a fixed noncentrality, the power of a χ^2 test decreases as df increase, so the directed test is more powerful, and the advantage grows with I and J.

9.5 Poisson regression for rates

Motivation. When events are counted over different amounts of exposure (time, space, population), the rate, not the raw count, is the natural response. A loglinear model with an offset models the rate while keeping the Poisson count as the observation.

9.5.1 Analyzing rates using loglinear models with offsets

Proposition 9.14 (Rate model with offset)

For counts n_i with exposure (index) t_i, the model

$$\log(\mu_i/t_i) = \alpha + \beta x_i \tag{9.7}$$

is the Poisson loglinear model $\log \mu_i = \log t_i + \alpha + \beta x_i$, in which $\log t_i$ is an offset (a predictor whose coefficient is fixed at 1). Equivalently $\mu_i = t_i \exp(\alpha + \beta x_i)$: the mean is proportional to the exposure, and e^β is the multiplicative effect of a unit increase in x on the rate.

Proof. Add $\log t_i$ to both sides of (9.7) and exponentiate. The ratio of rates at $x+1$ and x is $e^{\alpha+\beta(x+1)}/e^{\alpha+\beta x} = e^\beta$. ■

The identity link $\mu_i/t_i = \alpha + \beta x_i$, i.e. $\mu_i = \alpha t_i + \beta x_i t_i$, needs no offset: it is a Poisson GLM with identity link, predictors t_i and $x_i t_i$ and no intercept. Effects are additive rather than multiplicative; fitting may fail if negative fitted values arise.

9.5.2 Modeling death rates for heart valve operations

Example 9.15 (Heart valve replacement)

109 patients, classified by valve type and age; time at risk is the total follow-up (months) of all patients in the cell, those who died and those censored.

Age		Aortic	Mitral
< 55	Deaths	4	1
	Time at risk	1259	2082
	Death rate	0.0032	0.0005
55+	Deaths	7	9
	Time at risk	1417	1647
	Death rate	0.0049	0.0055

A binary model for the $2 \times 2 \times 2$ table (died yes/no) would be wrong, since patients had different exposures (3 to 97 months). With dummies a (age 55+) and v (mitral),

$$\log(\mu_{ij}/t_{ij}) = \alpha + \beta_1 a_i + \beta_2 v_j. \tag{9.8}$$

Estimates: $\hat{\beta}_1 = 1.221$ (SE 0.514), $\hat{\beta}_2 = -0.330$ (SE 0.438). Given valve type, the older group's death rate is $e^{1.221} = 3.4$ times the younger group's; 95% Wald CI $\exp(1.221 \pm 1.96 \times 0.514) = (1.2, 9.3)$, likelihood-ratio CI $(1.3, 10.4)$. Fit: $G^2 = 3.2$, $X^2 = 3.1$, df $= 1$; the model without valve type has $G^2 = 3.8$ (df 2); models without age fit poorly. Fitted death rates: < 55: 0.0018 (aortic), 0.0013 (mitral); 55+: 0.0062, 0.0044. The identity-link model fits with $G^2 = 1.1$ (df 1); its $\hat{\beta}_1 = 0.0040$ (SE 0.0014) is an additive difference in death rates. Only 21 of 109 patients died, so estimates are imprecise, but every patient contributes through the times at risk.

9.6 Empty cells and sparseness in modeling contingency tables

Motivation. Multiway tables quickly become sparse: many cells, many zeros. Zeros can make ML estimates infinite or nonexistent, and they spoil the χ^2 approximation for G^2 and X^2. We need to know when estimates exist and which statistics remain trustworthy.

9.6.1 Empty cells: sampling versus structural zeros

Definition (Sampling and structural zeros)

A *sampling zero* is a cell with $n_i = 0$ but $\mu_i > 0$; with larger n it would eventually be positive. A *structural zero* is a cell in which observations are impossible, $\mu_i = 0$ (e.g. prostate cancer among women). Tables with structural zeros are *incomplete*.

Sampling zeros are data and enter the likelihood; structural zeros are not data and are excluded from the model. The rest of this section concerns sampling zeros.

9.6.2 Existence of estimates in loglinear/logit models

Lemma 9.16 (Existence criterion)

Let $\mathcal{V} = \mathrm{col}(\mathbf{X})$ ($\mathbf{X}$ of full rank). Finite ML estimates exist if and only if there is no nonzero $\mathbf{d} \in \mathcal{V}$ with $d_i \le 0$ for all i and $d_i = 0$ whenever $n_i > 0$.

Proof. [Supplement]

Step 1 (necessity). Suppose such $\mathbf{d}$ exists, and let $\boldsymbol{\eta} \in \mathcal{V}$. Since $n_i = 0$ where $d_i < 0$,

$$L(\boldsymbol{\eta} + s\mathbf{d}) = \sum_{i:d_i=0} (n_i\eta_i - e^{\eta_i}) - \sum_{i:d_i<0} e^{\eta_i + sd_i},$$

which is strictly increasing in s (at least one $d_i < 0$). So no $\boldsymbol{\eta}$ is a maximizer.

Step 2 (sufficiency). Suppose no such $\mathbf{d}$ exists; we show $L(\boldsymbol{\eta}) \to -\infty$ as $\|\boldsymbol{\eta}\| \to \infty$ in $\mathcal{V}$. Otherwise there are $\boldsymbol{\eta}_m \in \mathcal{V}$ with $\|\boldsymbol{\eta}_m\| \to \infty$ and $L(\boldsymbol{\eta}_m) \ge c$. Pass to a subsequence with $\boldsymbol{\eta}_m/\|\boldsymbol{\eta}_m\| \to \mathbf{d} \in \mathcal{V}$, $\|\mathbf{d}\| = 1$. Each term $n_i\eta_i - e^{\eta_i}$ is bounded above (by $n_i \log n_i - n_i$, or 0 if $n_i = 0$), so each term is bounded below along the sequence.

Step 3. If $d_i > 0$, then $\eta_{m,i} \to +\infty$ and $n_i\eta_{m,i} - e^{\eta_{m,i}} \to -\infty$, impossible; so $d_i \le 0$. If $n_i > 0$ and $d_i < 0$, then $\eta_{m,i} \to -\infty$ and $n_i\eta_{m,i} \to -\infty$, impossible; so $d_i = 0$ whenever $n_i > 0$. Then $\mathbf{d}$ is a forbidden direction, a contradiction.

Step 4. Hence the superlevel sets of L on $\mathcal{V}$ are bounded and closed, L attains its maximum, and since $\mathbf{X}$ has full rank, $\hat{\boldsymbol{\beta}}$ is finite. ∎

Theorem 9.17 (Haberman's results)

For Poisson sampling:
(1) The log likelihood is a strictly concave function of $\log \boldsymbol{\mu}$.
(2) If an ML estimate of $\boldsymbol{\mu}$ exists, it is unique and satisfies $\mathbf{X}'\mathbf{n} = \mathbf{X}'\hat{\boldsymbol{\mu}}$; conversely, a $\boldsymbol{\mu}$ that satisfies the model and these equations is the ML estimate.
(3) If all $n_i > 0$, ML estimates of the parameters exist.
(4) If ML estimates exist for a model whose likelihood equations match certain marginal tables, those marginal tables have all counts positive.
(5) If ML estimates exist for a model M, they exist for every special case of M.
(6) The ML estimates $\hat{\boldsymbol{\mu}}$ are identical for multinomial and independent Poisson sampling and exist in the same situations.

Proof. [Supplement]

(1) and (2) are Theorem 8.19 (steps 1–3 of its proof).

(3) If all $n_i > 0$, a $\mathbf{d}$ with $d_i = 0$ whenever $n_i > 0$ is $\mathbf{0}$, so Lemma 9.16 gives existence.

(4) If the MLE exists, $\hat{\mu}_i = e^{\hat{\eta}_i} > 0$ for every cell, so all fitted marginal counts are positive, and by (2) they equal the observed marginal counts.

(5) A special case M_0 has $\mathcal{V}_0 \subseteq \mathcal{V}$. A forbidden direction in $\mathcal{V}_0$ would be a forbidden direction in $\mathcal{V}$; since none exists for M, none exists for M_0, and Lemma 9.16 applies.

(6) By Theorem 8.21, $L = [\text{multinomial log likelihood in } \boldsymbol{\beta}] + [n \log \tau - \tau]$, and the second bracket is maximized at the finite value $\tau = n$ for any data. So $\boldsymbol{\beta}$ has a finite maximizer under one scheme iff under the other, the maximizers coincide, and $\hat{\boldsymbol{\mu}} = n\boldsymbol{\pi}(\hat{\boldsymbol{\beta}})$ in both. ■

For the saturated model, $\hat{\boldsymbol{\mu}} = \mathbf{n}$ and the parameters (contrasts of $\log \hat{\mu}_i$) are finite only if all $n_i > 0$. For unsaturated models, estimates exist if all $n_i > 0$ and do not exist if a sufficient marginal count is 0. If some $n_i = 0$ but all sufficient margins are positive, Glonek et al. showed that estimates of a hierarchical model then always exist iff the model is decomposable (direct). For non-decomposable models like (XY, XZ, YZ) estimates exist when only one cell is empty but may fail with two:

Proposition 9.18 (A nonexistence example)

In a $2 \times 2 \times 2$ table with $n_{111} = n_{222} = 0$ and all other counts positive (all two-way margins positive), the ML estimates for (XY, XZ, YZ) do not exist.

Proof. [Supplement]

Step 1. For $2 \times 2 \times 2$ tables, $\mathcal{V}$ for (XY, XZ, YZ) has dimension 7, and the three-factor contrast $c_{ijk} = (-1)^{i+j+k}$ is orthogonal to it: for any $a_{ij} + b_{ik} + e_{jk}$, summing $(-1)^{i+j+k} a_{ij}$ over k gives 0, and similarly for the other terms. Hence $\mathcal{V} = \{\mathbf{d} : \mathbf{c}'\mathbf{d} = 0\}$.

Step 2. Let $d_{111} = d_{222} = -1$ and $d_{ijk} = 0$ otherwise. Then $\mathbf{c}'\mathbf{d} = -(-1)^3 - (-1)^6 = 1 - 1 = 0$, so $\mathbf{d} \in \mathcal{V}$; $\mathbf{d} \leq \mathbf{0}$, $\mathbf{d} \neq \mathbf{0}$, and $d_{ijk} = 0$ wherever $n_{ijk} > 0$.

Step 3. By Lemma 9.16, no finite ML estimate exists. ■

Extended ML estimates and infinite parameters

The supremum of the likelihood is always finite, so *extended* ML estimates of $\boldsymbol{\mu}$ exist, possibly with some $\hat{\mu}_i = 0$ on the boundary (e.g. $\hat{\mu}_i = 0$ when $n_i = 0$ in the saturated model). Then some parameter estimates are $\pm\infty$ and some odds ratios 0 or ∞. A zero XY marginal total gives infinite $\hat{\lambda}_{ij}^{XY}$ in (XY, XZ, YZ), and infinite effects of X in logit models for Y. Software is usually fooled by the nearly flat likelihood: it reports "convergence" with huge estimates

and enormous, unstable standard errors. Infinite estimates are not fatal: the likelihood-ratio confidence interval has one finite endpoint, e.g. $(-\infty, U)$ for $\log\theta$ when $n_{11} = 0$ in a 2×2 table. Zero patterns that force fitted zeros also change the df for testing fit.

9.6.3 Clinical trials example

Example 9.19 (Multicenter trial with zero successes)

Active drug versus placebo for fungal infections at five centers (Y: success/failure, X: treatment, Z: center).

Center	Treatment	Success	Failure	YZ margin	
1	Drug / Placebo	0 / 0	5 / 9	0	14
2	Drug / Placebo	1 / 0	12 / 10	1	22
3	Drug / Placebo	0 / 0	7 / 5	0	12
4	Drug / Placebo	6 / 2	3 / 6	8	9
5	Drug / Placebo	5 / 2	9 / 12	7	21
XY margin	Drug / Placebo	12 / 4	36 / 42		

Centers 1 and 3 have no successes, so the YZ margin has zeros, and for $\text{logit}\, P(Y = 1) = \beta x + \beta_k^Z$ (no intercept), $\hat\beta_1^Z = \hat\beta_3^Z = -\infty$. Software reports about -26 with SE about 200,000. The XY margin is positive and the treatment estimate is finite: $\hat\beta = 1.55$ (SE 0.70), the same as after deleting centers 1 and 3. Combining centers 1–3 gives $\hat\beta = 1.56$ (SE 0.70).

Proposition 9.20 (Uninformative strata)

In the logit model $\text{logit}\,\pi_{ik} = \beta x_i + \beta_k^Z$ with free center parameters, a center in which all responses are failures (or all successes) contributes nothing to the profile likelihood of β; hence $\hat\beta$ (and its likelihood-based inference) equals that obtained by deleting the center.

Proof. [Supplement] Center k with only failures contributes $\prod_i (1-\pi_{ik})^{n_{ik}}$, which for every fixed β increases to 1 as $\beta_k^Z \to -\infty$. Since β_k^Z appears in no other factor, the profile likelihood $\sup_{\{\beta_k^Z\}} \ell(\beta, \{\beta_k^Z\})$ equals the product over the remaining centers of their own profiled contributions, times 1. This is the profile likelihood with center k deleted, so it has the same maximizer and the same curvature. The all-success case is symmetric ($\beta_k^Z \to +\infty$). ■

Such strata also contribute nothing to the CMH test and the exact conditional test, although they do carry information about measures such as the difference of proportions.

9.6.4 Effect of small samples on X^2 and G^2

For fixed N cells, X^2 and G^2 are asymptotically χ^2 as $n \to \infty$; the quality depends on n and N. Cochran's rule for independence tests with df > 1: a minimum expected count of about 1 is acceptable if no more than about 20% of the $\hat\mu_i$ are below 5. X^2 is usable for smaller n and sparser tables than G^2; G^2 is usually poorly approximated when $n/N < 5$. When most $\hat\mu_i < 0.5$, G^2 is highly conservative (reported P-values too large); when most are between 0.5 and 4, it tends to be too liberal. For fixed n, N, tests with smaller df are better approximated: for conditional independence, $G^2[(XZ, YZ) \mid (XY, XZ, YZ)]$ (df $(I-1)(J-1)$) beats $G^2(XZ, YZ)$ (df $K(I-1)(J-1)$), and the ordinal df $= 1$ test of Section 9.4 is better still.

9.6.5 Model-based tests and sparseness

By (9.2)–(9.3), $G^2(M_0 \mid M_1)$ and $X^2(M_0 \mid M_1)$ depend on the data only through the sufficient statistics of M_1, usually marginal tables. Marginal totals are closer to normal than single cells, so these statistics approach χ^2 faster than $G^2(M_0)$ or $X^2(M_0)$. If the sufficient marginal totals of M_1 are mostly at least 5–10, the χ^2 approximation for model comparisons is usually adequate even when cell counts are small.

9.6.6 Alternative asymptotics and alternative statistics

When asymptotics fail, use exact small-sample methods, or Monte Carlo approximations of exact distributions. Alternatively, *sparse asymptotics* let N grow with n (cell means need not grow); then suitably standardized G^2 and X^2 are approximately normal (Koehler and Larntz; Koehler; McCullagh; Zelterman).

9.6.7 Adding constants to cells of a contingency table

Adding a small constant (e.g. $\frac{1}{2}$) to each cell guarantees finite estimates and convergence, and for saturated models it reduces bias in estimating an odds ratio. But for unsaturated models it oversmooths, distorting estimates and test statistics, especially with many cells. Even for a saturated model it replaces an honest infinite estimate (with interval (L, ∞)) by an arbitrary finite one. When in doubt, do a sensitivity analysis: delete or move possibly influential observations and see whether conclusions change. Random effects and Bayesian methods (later chapters) smooth in a less ad hoc way.